

Grant Swift

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Seeking a summer engineering internship.

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I like building things. I also like trying to break them. Most of my work goes into designing systems that hold up against anything I can throw at them, then testing until something gives, and rebuilding it stronger each time.

SKILLS

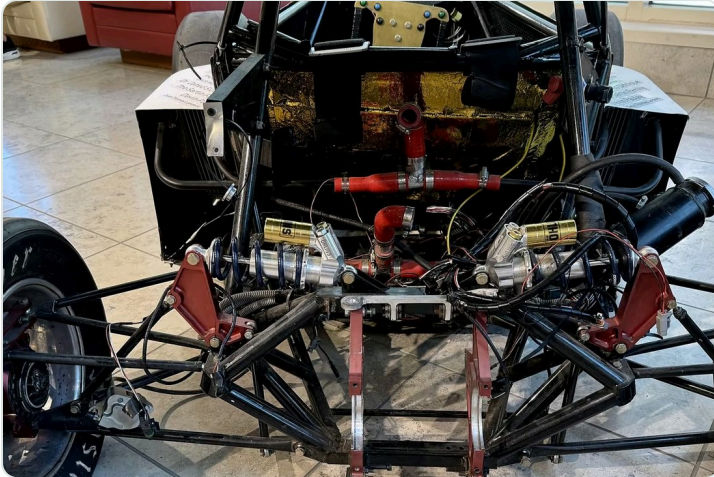
CAD & Design	SolidWorks, AutoCAD, GD&T and blueprint reading, tolerance analysis
Analysis	Statics, Dynamics, Thermodynamics, Design & Manufacturing Processes
Manufacturing	CNC and manual machining, inspection and measurement, 3D printing, lathe work
Software	JavaScript, Node.js, Express, React, SQL, REST API design
Teams	OU Formula SAE — Sooner Racing, powertrain subteam

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Formula SAE Cooling Package

Sized, packaged and mounted a temporary radiator for the Sooner Racing car so the team could start on-track testing **about a month before** the final cooling package existed.



The chassis the radiator had to live in. It was not designed around it.



Running trim. Temporary package, real track time.



Package. Front structure and the tubes the mounts tie into.



Build. Fabrication and fit-up in the shop.

THE THREE DECISIONS

FRONTAL AREA OVER CORE DEPTH

Testing is low speed, so airflow is the scarce thing. A deeper core mostly adds pressure drop to air you are already struggling to move. That put it at roughly **12 by 18 in**, hex-cell.

12 × 18 in

core face, approx.

THE CORE IS NOT A CHASSIS MEMBER

A radiator core is thin-wall brazed material. I designed a support frame that carries the load and captures the core, so nothing structural passes through the radiator itself.

Hex-cell

core matrix

TEMPORARY MEANS REMOVABLE

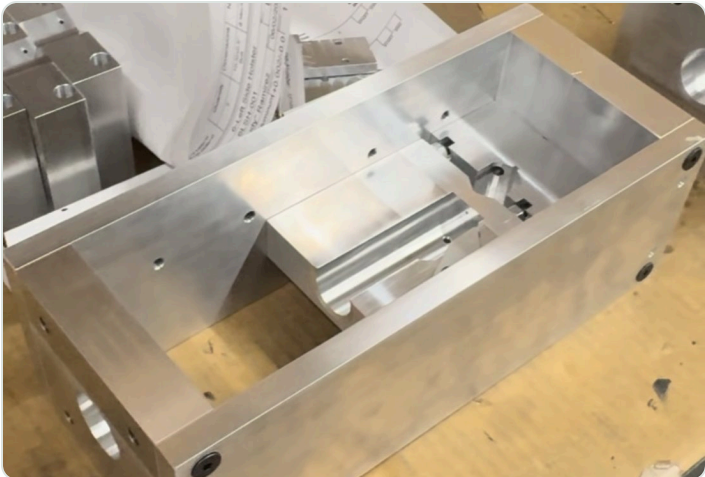
Brackets tie to chassis tubes already on the car. No permanent modification to a frame the real cooling package still has to use, and it comes off in a few fasteners.

0

permanent chassis mods

Tolerance Stack-Up on a Client Assembly

Two summers machining, measuring and troubleshooting parts for a client build. My main job was a run of aluminum brackets that had to pass inspection *and* actually go together in the finished assembly.



The assembly the machined parts had to fit into.



Checking a finished bracket against the print.



The part. Client drawing, revision E, held to ± 0.005 in.



Shop piece. Turned and engraved on the lathe.

WHAT I WORKED OUT

IN SPEC IS NOT THE SAME AS FITS

Parts could each pass their own dimensions and still refuse to go together. The problem lived in how the tolerances stacked at the interfaces, not in any one number.

MEASURE BEFORE YOU MACHINE MORE

I inspected parts against the drawing with calipers as they came off, so a drifting dimension got caught on one part instead of a whole batch.

THE CLIENT'S PRINT IS THE SPEC

It had to be exactly what was drawn and it had to work assembled. Anything I changed to make parts fit still had to land inside the drawing.

2
summers at SAGA

± 0.005 in
drawing tolerance

Rev E
client drawing revision

Aluminum
material

PROJECT 03

3D Printed Paddle Shifters

Extended paddle shifters for my Audi A7, designed from scratch to fit the factory mounts. Modelled, printed, test fitted, refined, then sanded and clear coated.

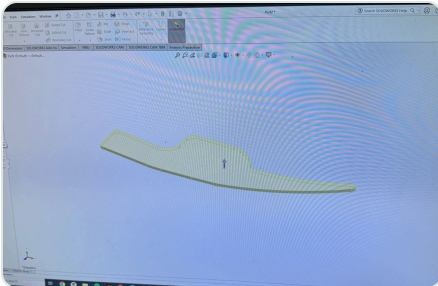


Original paddle mount.

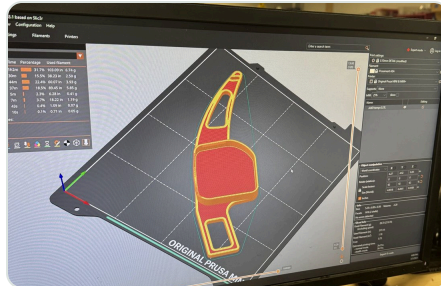


Paddles after sanding and clear coat.

PROCESS



1 • Model. Surfaced the paddle profile in SolidWorks.



2 • Print. Orientation and supports set for strength.



3 • Final look. Both paddles fitted, circled.

WHAT MADE IT HARD

FIT BEFORE FORM

The mount had to grip the factory paddle exactly. I printed fit checks before committing to a full part, so the tolerance was proven early.

SURVIVED A TEXAS SUMMER

PLA starts softening near 140°F, and a parked car in 105°F sun runs past 130°F. The paddles have held their shape and fit with no warping.

FINISH IT PROPERLY

Sanded through grits and clear coated so the part reads as a factory interior piece instead of a printed one.

6

design iterations

5

test prints

PLA

filament

3 h

total print time

Woodworking: Handcrafted Pens

Pens built from scratch on the lathe. Raw blanks get drilled, sleeved with brass, turned to size, finished and press fit to hardware. An ongoing exercise in doing the same precise thing repeatably, by hand.



Finished pens across several kits and blank materials.



One production run, turned to match.



Where they start

Blanks are drilled and a brass tube is epoxied inside, then sorted by kit. That tube is the reference every later operation works from, so getting it straight decides whether the finished pen is square.

WHAT THE WORK DEMANDS

EVERY PIECE IS UNIQUE

No two blanks have the same grain, so no two pens finish the same. Reading the wood and orienting it before it ever hits the lathe is half the design.

PRESS FIT, NO FASTENERS

Components press into the tubes on an interference fit. Too loose and it spins, too tight and the blank splits. The ends have to be square and to length.

IDENTICAL WHERE IT COUNTS

The grain is always different, the geometry never is. Diameter, sanding sequence and finish stay the same on every piece so each one still fits its hardware.

Dozen+

styles

1000+

pens handcrafted

Wood & acrylic

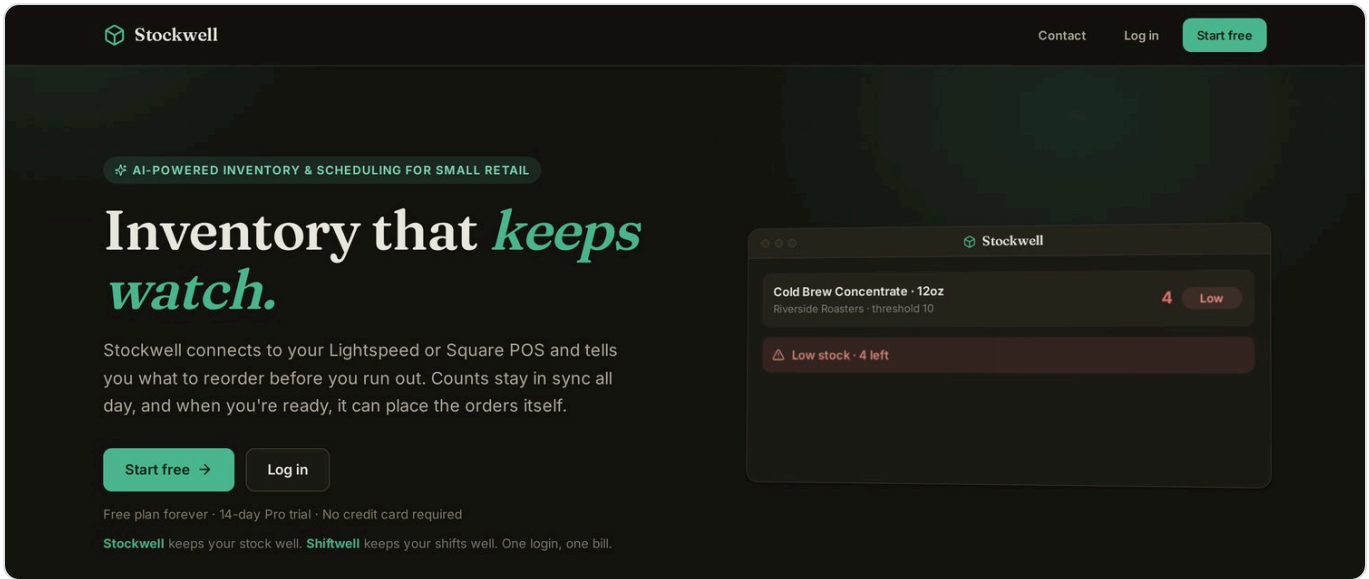
blank materials

Ongoing

still making them

Stockwell

Software that saves small retail shops **time** and **money**. It schedules the staff and restocks the shelves. Built and shipped solo.



SAVES TIME · SCHEDULING

Staff availability goes in, a finished weekly schedule comes out. What took a manager hours of rearranging every week now takes a review and a click.

SAVES MONEY · RESTOCKING

It watches stock levels in real time and reorders from the vendor before anything runs out, so shelves stay full and nothing sells out at full price.

INTEGRATES WITH

- Lightspeed Retail
- Lightspeed X-Series
- Square
- Stripe
- Claude API
- Resend
- SendGrid

ENGINEERING HIGHLIGHTS

NEVER TRUST THE AI

The AI proposes a schedule; separate code verifies it. Double-bookings and untrained assignments get repaired, not published.

PROVEN ISOLATION

Many businesses, one server. A test battery has one try to reach another's data on every run. Every attempt must fail.

BUILT TO BE SWAPPED

Database, AI, email and imports each sit behind small interfaces, so any one can be replaced without touching the rest.

~29,000
lines across 156 files

300+
automated tests

3
POS integrations

24
database tables